

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 4

For the tutorial in Week 5 on 15 February, let us discuss

- Ex. 6.5.1, 6.5.4, 6.5.12, 6.5.11, 6.5.14, 6.5.19, 6.5.32, 8.1.10, additional exercises.

Ex. 6.5.11. How many solutions are there to the equation $x_1 + x_2 + x_3 + x_4 + x_5 = 21$, where x_i , $i = 1, 2, 3, 4, 5$, is a non-negative integer, such that

- (a) $x_1 \geq 1$?
- (b) $x_i \geq 2$ for $i = 1, 2, 3, 4, 5$?
- (c) $0 \leq x_1 \leq 10$?
- (d) $0 \leq x_1 \leq 3$ and $x_2 \geq 15$?

Solution:

- (a) Define $x_1 = 1 + y_1$, so the equation is equivalent to $y_1 + x_2 + x_3 + x_4 + x_5 = 20$. Hence, there are $C(5 + 20 - 1, 20) = C(24, 4) = 10626$ solutions.
- (b) Define $x_i = 2 + y_i$, so the equation is equivalent to $y_1 + y_2 + y_3 + y_4 + y_5 = 11$. Hence, there are $C(5 + 11 - 1, 11) = C(15, 4) = 1365$ solutions.
- (c) (i) The total number of solutions without any restrictions is $C(5 + 21 - 1, 21) = C(25, 4)$.
(ii) The total number of solutions with $x_1 \geq 11$ is $C(5 + 10 - 1, 10) = C(14, 4)$. Hence, the number of solutions with $0 \leq x_1 \leq 10$ is $C(25, 4) - C(14, 4) = 11649$.
- (d) (i) The total number of solutions with $x_2 \geq 15$ is $C(5 + 6 - 1, 6) = C(10, 4)$. (ii) The total number of solutions with $x_1 \geq 4$ and $x_2 \geq 15$ is $C(5 + 2 - 1, 2) = C(6, 2)$. Hence, the number of solutions with $0 \leq x_1 \leq 3$ and $x_2 \geq 15$ is $C(10, 4) - C(6, 2) = 195$.

Ex. 6.5.14. How many solutions are there to the inequality

$$x_1 + x_2 + x_3 \leq 11,$$

where x_1, x_2, x_3 are non-negative integers?

[Hint: introduce an auxiliary variable x_4 such that $x_1 + x_2 + x_3 + x_4 = 11$.]

Solution: $x_1 + x_2 + x_3 \leq 11$ if and only if $x_1 + x_2 + x_3 + x_4 = 11$ for some integer $x_4 \geq 0$. There are $C(11 + 4 - 1, 11) = C(14, 3)$ solutions for the latter.

Ex. 6.5.19. How many positive integers strictly lesser than 1,000,000 have the sum of their digits equal to 19 ?

Solution: The positive integers range from 1 to 999,999 and so we may consider each integer as a string of 6 digits (including those where 0's appear in front). Let us denote the digits in the string as $x_1, x_2, x_3, x_4, x_5, x_6$. Then

$$\sum_{i=1}^6 x_i = 19, \quad \text{where } 0 \leq x_i \leq 9.$$

There are $C(19 + 6 - 1, 19) = C(24, 5)$ solutions if there are no restrictions on the range of x_i 's.

We need then, to subtract away those cases where some x_i 's are outside of the range, i.e., where $x_i \geq 10$. Note that this can only happen to at most one out of the 6 variables, for otherwise if two or more of the x_i 's are greater than 9, then $\sum_{i=1}^6 x_i$ would exceed 19.

Suppose x_j is the variable that is out of range, i.e., $x_j \geq 10$, while the other variables are within range. Then the number of solutions where

$$\sum_{i=1}^6 x_i = 19, \quad \text{where some } x_j \geq 10, \text{ and } 0 \leq x_i \leq 9 \text{ for } i \neq j,$$

is $C(19 - 10 + 5, 5) = C(14, 5)$. We multiply this number by 6 since there are 6 choices for j . Therefore the final answer is

$$C(24, 5) - 6 \cdot C(14, 5) = 30492.$$

Ex. 6.5.1, 6.5.4, additional exercise.

- (a) In how many different ways can five elements be selected **in order** from a set with three elements when repetition is allowed ?
- (b) How many ways are there to select five unordered elements from a set with three elements when repetition is allowed ?
- (c) How many ways are there to assign three distinct jobs to five (of course distinct) employees if each employee can be given more than one jobs ?

Solution:

- (a) 3^5 . Of the five elements to be selected, each has 3 choices, hence 3^5 in total by product rule.
- (b) This is 5-combinations of 3 types, i.e., $C(n + r - 1, r) = C(7, 5) = C(7, 2) = 21$ with $r = 5, n = 3$.
- (c) 5^3 . Of each of the 3 jobs, there are 5 choices to whom it can be assigned, hence 5^3 in total by product rule.

Ex. 6.5.12, additional exercise.

- (a) How many strings of 10 ternary digits (0, 1, or 2) are there that contain exactly two 0's, three 1's, and five 2's ?
- (b) Let n be a multiple of k . How many ways can n books be placed on k distinguishable shelves if no two books are the same, the positions of the books on the shelves do not matter, and each shelf has exactly n/k books?

Solution:

- (a) By applying theorem from lecture directly (Section 6.5, Theorem 3), the number of ways is

$$\frac{10!}{2!3!5!} = 2520.$$

- (b) By a direct application of theorem from lecture (Section 6.5, Theorem 4), the number of ways is

$$\frac{n!}{(n/k)!(n/k)!\cdots(n/k)!} = \frac{n!}{((n/k)!)^k}.$$

Ex. 6.5.32 modified. How many ways can n books be placed on k distinguishable shelves

- (a) if the books are indistinguishable copies of the same title?
 (b) if no two books are the same, and the positions of the books on the shelves matter?

Solution:

- (a) Consider each shelf as a different “element” in a set S , and placing a book on a shelf means selecting a copy of this element. Then the problem is equivalent of asking the number of n -combinations with repetition allowed from a set of k elements.

So the answer is $C(k + n - 1, n)$.

- (b) Consider this problem in two steps: (1) find a placement of the books on k shelves as if the books are all indistinguishable. From the previous case we know there are $C(k + n - 1, n)$ ways; (2) make the books distinguishable by giving them an order on the shelves. Note there are always $n!$ possible orders of a sequence of n books, regardless of their placement on the shelves.

By the product rule, the final answer should be the product of these two values, which is $n! \cdot C(k + n - 1, n)$.

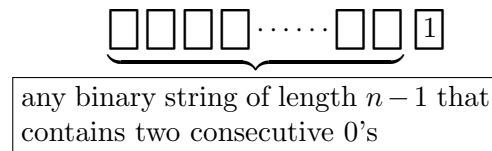
Ex. 8.1.10. A binary string is a string containing only 0's and 1's.

- (a) find a recurrence relation for the number of bit strings of length n that contain two consecutive 0's.
 (b) what are the initial conditions ?
 (c) find the answer when $n = 4$.

Solution:

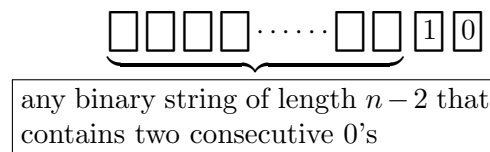
Let a_n be the number of such ternary strings of length n .

- (a) **Case 1:** Consider the strings when the last bit is 1. Number of ternary strings = a_{n-1}

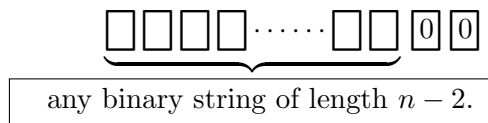


Next, we consider the strings where the last bit is 0. In this case, we need to consider the penultimate bit too.

Case 2: Last two bits are 10 then number of strings is a_{n-2} .



Case 3: Last two bits are 00. Then there are 2^{n-2} strings, since any binary string of length $n - 2$ with the additional 00 bits at the end would satisfy the required condition.



So in total, we have $a_n = a_{n-1} + a_{n-2} + 2^{n-2}$, for $n \geq 3$.

- (b) $a_1 = 0, a_2 = 1$.
- (c) $a_4 = 2^2 + 2a_2 + 2^1 + a_1 = 4 + 2 + 2 + 0 = 8$. (They are 0000, 0001, 0010, 0011, 0100, 1100, 1000, and 1001.)

Additional Exercise on coffee. Bottle T has 4 litres of tea and bottle C has 4 litres of coffee. In each step you pour 1 litre of liquid from bottle T into bottle C, stir, pour back 1 litre of liquid from bottle C into bottle T, and stir again. (This is a recipe for Kopi Cham, see https://en.wikipedia.org/wiki/Kopi_tiam.) Let $f(n)$ be the amount of coffee in bottle T after n steps.

- (a) Calculate the values $f(0)$, $f(1)$, and $f(2)$.
- (b) Write down the recurrence formula for $f(n)$.
- (c) Solve the recurrence from part (2).
- (d) What is the limiting value of $f(n)$ as n approaches infinity?

Solution:

- (a) Initially, bottle T only contains 4 litres of tea, so $f(0) = 0$.
In 1st step, we pour 1 litre of tea from bottle T to bottle C. So, bottle C has 1 litre of tea and 4 litres of coffee. After stirring, we pour 1 litre of liquid, including $4/5$ of coffee, from bottle C to bottle T. Therefore, bottle T has $4/5$ of coffee and so $f(1) = 4/5$. Also, bottle C has $4 - 4/5 = 16/5$ litres of coffee.
In 2nd step, we pour 1 litre of liquid, including $1/4 \cdot 4/5 = 1/5$ litres of coffee from bottle T to bottle C. So, bottle C has 5 litres of liquid, including $16/5 + 1/5 = 17/5$ litres of coffee. After stirring, we pour 1 litre of liquid, including $(17/5)/5 = 17/25$ litres of coffee, from bottle C to bottle T. Therefore, bottle T has $4/5 - 1/5 + 17/25 = 32/25$ litres of coffee and so $f(2) = 32/25$.
- (b) Note that both bottles remain 4 litres of liquid after each step. After n steps, bottle T has $f(n)$ litres of coffee and bottle C has $4 - f(n)$ litres of coffee, since the total amount of coffee must be 4.
We start $(n + 1)$ th step. We first pour 1 litre of liquid, containing $1 \cdot \frac{f(n)}{4}$ litres of coffee, from bottle T to bottle C. Then, bottle C has 5 litres of liquid, including $4 - f(n) + \frac{1}{4}f(n) = 4 - \frac{3}{4}f(n)$ litres of coffee. Next we pour 1 litre of liquid, including $1 \cdot (4 - \frac{3}{4}f(n))/5 = \frac{4}{5} - \frac{3}{20}f(n)$ litres of coffee, from bottle C to bottle T. After that bottle T contains $f(n) - \frac{1}{4}f(n) + (\frac{4}{5} - \frac{3}{20}f(n))$ litres of coffee, that is

$$f(n+1) = f(n) - \frac{1}{4}f(n) + \left(\frac{4}{5} - \frac{3}{20}f(n)\right) = \frac{4}{5} + \frac{3}{5}f(n), \quad \text{for } n \geq 0.$$

(c) We try to guess a solution by unwinding the formula for $f(n)$:

$$\begin{aligned} f(n) &= \frac{4}{5} + \frac{3}{5}f(n-1) \\ &= \frac{4}{5} + \left(\frac{3}{5}\right) \cdot \frac{4}{5} + \left(\frac{3}{5}\right)^2 f(n-2) \\ &= \frac{4}{5} + \left(\frac{3}{5}\right) \cdot \frac{4}{5} + \left(\frac{3}{5}\right)^2 \cdot \frac{4}{5} + \left(\frac{3}{5}\right)^3 f(n-3) \end{aligned}$$

Continuing in this manner suggests the guess

$$f(n) = \frac{4}{5} + \left(\frac{3}{5}\right) \cdot \frac{4}{5} + \left(\frac{3}{5}\right)^2 \cdot \frac{4}{5} + \cdots + \left(\frac{3}{5}\right)^{n-1} \cdot \frac{4}{5} + \left(\frac{3}{5}\right)^n \cdot f(0)$$

Since $f(0) = 0$, we can write

$$\begin{aligned} f(n) &= \frac{4}{5} + \left(\frac{3}{5}\right) \cdot \frac{4}{5} + \left(\frac{3}{5}\right)^2 \cdot \frac{4}{5} + \cdots + \left(\frac{3}{5}\right)^{n-1} \cdot \frac{4}{5} \\ &= \frac{\frac{4}{5} \left(1 - \left(\frac{3}{5}\right)^n\right)}{1 - \frac{3}{5}} \\ &= 2 - 2\left(\frac{3}{5}\right)^n \end{aligned}$$

We prove this is correct by induction on n . When $n = 0$, both $f(0)$ and $2 - 2\left(\frac{3}{5}\right)^n$ are 0. Now assume $f(n) = 2 - 2\left(\frac{3}{5}\right)^n$ for some n . Then

$$f(n+1) = \frac{4}{5} + \frac{3}{5}f(n) = \frac{4}{5} + \frac{3}{5} \left(2 - 2\left(\frac{3}{5}\right)^n\right) = \frac{4}{5} + \frac{6}{5} - \frac{6}{5} \left(\frac{3}{5}\right)^n = 2 - 2\left(\frac{3}{5}\right)^{n+1}$$

as it should be.

(d) Since $\left|\frac{3}{5}\right| < 1$, it is easy to see $\lim_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} \left(2 - 2\left(\frac{3}{5}\right)^n\right) = 2$.

Hints.

Ex. 6.5.32(b). The number of ways can be computed in a two step process. Step 1: Count how many ways there are to distribute the n books into k shelves as if the n books are indistinguishable and the order on the shelves do not matter. Step 2: Count how many permutations there are of the n books, this is to take care of the fact that the books are distinguishable and the order of the books on the shelves matter. Finally, apply the product rule.

Ex. 8.1.10(a). Just like the lecture examples, consider the last bit. For the case when the last bit is 0, we split into two cases, whether the last two bits are 10 or 00.

Additional Exercise on coffee This question is not difficult, just long. Note that when there is $f(n)$ amount of coffee in bottle T, it means there is $4 - f(n)$ amount of coffee in bottle C.